

Hospital Management System

A DBMS Capstone Project Report submitted in partial fulfillment of the requirements for the award
of the degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted By

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

ADITYA UNIVERSITY

(Formerly Aditya Engineering College (A))

Aditya Nagar, ADB Road, Surampalem, Andhra Pradesh, India

2026 – 2027

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CERTIFICATE

This is to certify that the DBMS Capstone Project report entitled “**Hospital Management System**” is being submitted by **D.R.PALGUN** (Roll No. **25B11AI241**), **K.ASHOK** (Roll No.**25B11AI478**), **P.V.S ROHITH REDDY** (Roll No. **25B11AI892**) and **G.M.SRIRAM** (Roll No. **25B11AI311**) in partial fulfillment of the requirements for the award of the B.Tech. degree in Artificial Intelligence and Machine Learning. The work documented in this report was carried out as part of the **Capstone Project** for the course **Database Management Systems (2501IT05)** during the academic year **2026–2027**, and records the database design, implementation, and documentation undertaken by the team.

Project Guide

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DECLARATION

We hereby declare that the DBMS Capstone Project entitled “**Hospital Management System**” is a genuine work carried out by us. The database design, SQL implementation, queries and documentation presented in this report are our own work, completed under the guidance of, **T.Srinivasulu** sir, Department of **Artificial Intelligence and Machine Learning**, Aditya University, Surampalem. We further declare that this work has not been submitted, in full or in part, for the award of any degree of this or any other educational institution.

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ACKNOWLEDGEMENT

It is with immense pleasure that we express our indebted gratitude to our project guide T.SRINIVASULU sir, Department of Artificial Intelligence & Machine Learning, who guided us throughout the capstone project. We are thankful for the constant encouragement and valuable suggestions received at every review stage.

We are grateful to Dr.K.NagaBhargavi, Head of the Department, for inspiring us and for arranging the facilities and resources needed for this project.

We extend our sincere thanks to the Management, Dr. M. Sreenivasa Reddy, Deputy Pro-Chancellor, Dr. M. B. Srinivas, Vice-Chancellor, and Dr. A. Ramesh, Pro Vice-Chancellor (Engineering & Sciences), for providing an excellent academic environment.

Not to forget, we thank the Teaching and Non-Teaching Staff, our parents and our friends, who have directly or indirectly supported us in completing this project on time.

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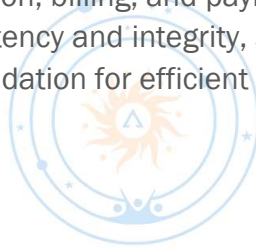
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ABSTRACT

The **Hospital Management System** is a database project developed to efficiently manage the day-to-day information and operations of a hospital. Hospitals handle large amounts of data related to patients, doctors, departments, appointments, treatments, admissions, medicines, prescriptions, billing, and payments. Managing such information using manual records or spreadsheets can result in data redundancy, inconsistency, difficulty in retrieval, and poor data integrity. The purpose of this project is to develop a centralized relational database that provides accurate storage, easy retrieval, and effective management of hospital information. The database consists of 15 entities, with major tables including **Department, Doctor, Patient, Appointment, Treatment, Ward, Room, Admission, Medical_Record, Prescription, Medicine, Bill, and Payment**. The system is implemented using MySQL and SQL-based database tools. SQL operations such as CREATE, INSERT, UPDATE, SELECT, WHERE, ORDER BY, aggregate functions, primary keys, foreign keys, UNIQUE constraints, and CHECK constraints are implemented. The database supports patient registration, doctor and department management, treatment records, prescriptions, room allocation, billing, and payments. The resulting system reduces data redundancy, improves consistency and integrity, simplifies information retrieval, and provides an organized foundation for efficient hospital data management.



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INDEX

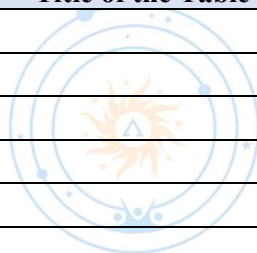
Chapter	Title	Page No.
1	Introduction	
	1.1 Introduction to the Domain	
	1.2 Problem Statement	
	1.3 Motivation and Need for a Database	
	1.4 Objectives of the Project	
	1.5 Scope of the Project	
	1.6 Organization of the Report	
2	System Study and Tools	
	2.1 Existing System and Its Limitations	
	2.2 Proposed Database System	
	2.3 Tools and Technologies Used	
	2.4 Software and Hardware Requirements	
3	Database Design (Review – I)	
	3.1 Identified Entities	
	3.2 Entity Attributes and Attribute Types	
	3.3 ER Diagram	
	3.4 Relationships and Cardinality	
	3.5 Relational Database Schema	
	3.6 Primary Keys and Foreign Keys	
	3.7 Normalization up to 3NF	
4	Database Implementation (Review – II)	
	4.1 Database and Table Creation – DDL	
	4.2 Sample Data Insertion – DML	
	4.3 Constraints and Data Integrity	
	4.4 Data Validation Rules	
5	SQL Operations and Queries	
	5.1 DQL – Basic Retrieval Queries	
	5.2 Join Operations	
	5.3 Aggregate Functions, GROUP BY and HAVING	
	5.4 Nested and Correlated Subqueries	
	5.5 Views and Relational Set Operations	
	5.6 DCL and TCL Operations	
6	Modules and Functionalities	
7	Results – Sample Outputs and Screenshots	
8	Testing and Data Validation	
9	GitHub Repository and Team Contribution	
10	Conclusion and Future Enhancements	
	References	
	Appendix A – Complete SQL Script	

LIST OF FIGURES

Figure No.	Title of the Figure	Page No.

LIST OF TABLES

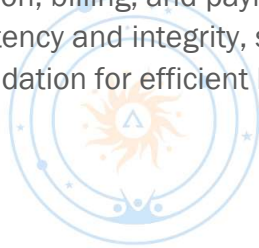
Table No.	Title of the Table	Page No.



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INDEX

Chapter	Title	Page No.
1	Introduction	
	1.1 Introduction to the Domain	
	1.2 Problem Statement	
	1.3 Motivation and Need for a Database	
	1.4 Objectives of the Project	
	1.5 Scope of the Project	
	1.6 Organization of the Report	
2	System Study and Tools	
	2.1 Existing System and Its Limitations	
	2.2 Proposed Database System	
	2.3 Tools and Technologies Used	
	2.4 Software and Hardware Requirements	
3	Database Design (Review – I)	
	3.1 Identified Entities	
	3.2 Entity Attributes and Attribute Types	
	3.3 ER Diagram	
	3.4 Relationships and Cardinality	
	3.5 Relational Database Schema	
	3.6 Primary Keys and Foreign Keys	
	3.7 Normalization up to 3NF	
4	Database Implementation (Review – II)	
	4.1 Database and Table Creation – DDL	
	4.2 Sample Data Insertion – DML	
	4.3 Constraints and Data Integrity	
	4.4 Data Validation Rules	
5	SQL Operations and Queries	
	5.1 DQL – Basic Retrieval Queries	
	5.2 Join Operations	
	5.3 Aggregate Functions, GROUP BY and HAVING	
	5.4 Nested and Correlated Subqueries	
	5.5 Views and Relational Set Operations	
	5.6 DCL and TCL Operations	
6	Modules and Functionalities	
7	Results – Sample Outputs and Screenshots	
8	Testing and Data Validation	
9	GitHub Repository and Team Contribution	
10	Conclusion and Future Enhancements	
	References	
	Appendix A – Complete SQL Script	

LIST OF FIGURES

Figure No.	Title of the Figure	Page No.
Fig. 3.1	ER Diagram of the Hospital Management System	10
Fig. 3.2	Relational Schema Diagram	15
Fig. 4.1	Successful Creation of Tables	20
Fig. 4.2	Sample Data Inserted into the Tables	22
Fig. 4.3	Referential Integrity Enforced by the Database	23
Fig. 7.1	Doctor whose experience greater than 7 years	30
Fig. 7.2	Medicines price greater than 30	30
Fig. 7.3	Average Experience	31
Fig. 7.4	List of all the treatments	31

LIST OF TABLES

Table No.	Title of the Table	Page No.
Table 2.1	Tools and Technologies Used	6
Table 2.2	Software and Hardware Requirements	6
Table 3.1	Entities Identified for the Selected Domain	7
Table 3.2	Attributes of Each Entity	7
Table 3.3	Relationships and Cardinality	11
Table 3.4	Relational Database Schema	11
Table 3.5	Primary Key and Foreign Key Definitions	15
Table 3.6	Normalization Steps	16
Table 4.1	Sample Data Summary	20
Table 4.2	Constraints and Data Integrity Rules	22
Table 5.1	Basic Retrieval Queries	24
Table 5.2	Join Operations	25
Table 5.4	Subqueries	26
Table 5.5	Views and Set Operations	26
Table 5.6	DCL and TCL Operations	27
Table 6.1	Modules and Their Functionalities	29
Table 8.1	Test Cases and Results	32
Table 9.1	Team Responsibilities	34
Table 9.2	Review-wise Progress Record	34